

7.1 Integration by Parts

How can we address multiplication in an integral such as $\int f(x)g(x) \, dx$?

But how would we calculate $\int xe^x \, dx$?

Deriving the Integration by Parts Formula

In this section, we learn how to integrate functions of the form

$$\int f(x)g(x) \, dx,$$

where $f(x)$ is a function of which we can easily take the _____,
while $g(x)$ can be _____ repeatedly.

Integration by Parts

Let $u = f(x)$ and $v = g(x)$. Then we can rewrite our formula as

Example 1

$$\int x e^x \, dx$$

How do we choose which function to use for u ?

Recall that u is a function of which we can easily derive, but not necessarily integrate. Here is a way to discern potential functions for u :

I

L

A

T

E

Example 2

$$\int x \sin(x) \, dx$$

What if there are not two functions in the integral?

Example 3

$$\int \ln(x) \, dx$$

Extra Example

$$\int \sin^{-1}(5y) \, dy$$

Sometimes we must use integration by parts multiple times.

Example 4

$$\int x^2 e^x \, dx$$

Example 5

$$\int e^x \sin(3x) \, dx$$